

INSTALLATION CASE STUDY

Air Source Heat Pump Cascade | Space Heating, Hot Water & Swimming Pool

Clean Energy. Smarter Future.



Two Heat Pumps. Three Jobs.

A high heat loss property, a full domestic hot water demand and a swimming pool to warm - solved with one intelligently split cascade rather than three separate systems.

SYSTEM

2 x ASHP cascade

CONFIGURATION

Shared duty

SERVES

Heating, DHW & Pool

INSTALLER

Elixia Renewables

THE BRIEF

A property that asked three things of one system

Most heat pump projects have one job to do: keep the house warm. This one had three - and each of them pulled the design in a different direction.

The property carried a **heat loss well above the 8-12 kW that covers the majority of UK homes** - already at the upper end of what a single domestic monobloc delivers comfortably in mid-winter. Add a full stored hot water demand across multiple bathrooms, and then a **swimming pool** asking for a long, low-grade, high-volume heat input, and a single appliance solution stops being ambitious and starts being fragile.

The problem with the obvious answers

One oversized heat pump. Size a single unit for peak heat loss plus pool and you end up with a machine that spends the other nine months of the year cycling badly against a fraction of its output. Short cycling is the fastest way to destroy both efficiency and compressor life.

Three separate systems. A heating unit, a hot water unit and a dedicated pool heater is clean on paper and expensive everywhere else - three sets of groundworks, three electrical supplies, three commissioning visits, three service contracts, and a plant area that no longer fits the space available.

The design question was not *how much heat do we need*. It was *how do we make the same kilowatts do different work at different times of year, without ever fighting each other*.

The design envelope

Indicative parameters for a cascade specification of this type. These are typical ranges for this class of system, not measured values for this property.

Property	Large detached, approx. 250-350 m ²
Design heat loss	18-26 kW at -2 to -4°C external
Emitters	35-40°C flow underfloor, 45-50°C radiators
Hot water	300-500 litre unvented cylinder, 3 kW immersion back-up
Pool	45-55 m ³ ; 4-8 kW steady demand with a cover fitted
Electrical supply	25-30 A per unit - three phase common at this scale

A pool cover is the single biggest variable above: an uncovered outdoor pool can demand three to five times the heat input of a covered one.

THE SOLUTION

A cascade, with one unit doing double duty

Rather than one large appliance or three small ones, we installed **two air source heat pumps in cascade**, sharing a common primary circuit - but with the two units given deliberately different roles.

1 Unit 1 - dedicated space heating

The lead unit is committed to the heating circuit. It carries the base load on its own for the majority of the year, running long, low, weather-compensated cycles at the lowest flow temperature the emitters will allow. No competing demand, no interruptions for hot water, no compromise on efficiency.

2 Unit 2 - shared duty across heating, hot water and pool

The second unit is the flexible one. Its output is split at the header so that it can support space heating alongside Unit 1 when outside temperatures fall, while also serving the domestic hot water cylinder and the pool heat exchanger when those loads call. In practice it behaves like two half-machines pointing in different directions.

3 Hydraulic separation as the referee

Both units feed a hydraulically separated primary, keeping the heat pump circuit independent of the distribution circuits. Every circuit - heating zones, cylinder coil, pool plate heat exchanger - draws through its own pump and isolation, so flow rates stay correct whichever combination of loads is running.

4 Controls that set the priority order

The controls decide who gets the heat and when. Hot water takes priority for short, defined windows. Space heating takes precedence over the pool during cold weather. The pool picks up surplus capacity in the shoulder seasons, when there is plenty of it going spare.

In mid-winter the property has the full output of two heat pumps behind the heating. In summer, one unit covers hot water while the other quietly heats the pool. Nothing is oversized for the job it is doing at the time.

Why the cascade wins

Turndown

Two smaller units modulate far lower than one large one, so the system matches part load instead of cycling against it.

Resilience

If one unit is offline for service or fault, the property still has heat and hot water. A single appliance gives you nothing.

Pool without penalty

Pool load is met from spare cascade capacity rather than a fourth appliance and a fourth running cost.

One plant room

Shared header, shared controls, shared electrical infrastructure, one commissioning process, one service visit.

THE INSTALLATION

Built once, built properly

Both units sit on level bases against the gable elevation, on anti-vibration mounts, with the primary flow and return runs kept short, insulated and neatly aligned between the two machines. Positioning was chosen for clear airflow across both fan banks and for acoustic performance at the nearest neighbouring window, assessed under MCS 020.

Internally the plant is arranged around the separated primary, with individually valved circuits so any zone, the cylinder or the pool exchanger can be isolated without draining the system.

Scope of works

- 2 x air source heat pumps installed in cascade configuration
- Hydraulic separation, primary pump set and fully valved circuit manifolds
- Unvented hot water cylinder with immersion back-up
- Plate heat exchanger and secondary circuit serving the swimming pool
- Weather compensation and priority-based control strategy
- Dedicated electrical circuits, isolation and protection
- System flushing, inhibitor dosing, commissioning and handover

Typical performance for a system of this type

3.0-3.5

Seasonal efficiency for a well-designed low flow temperature cascade

20-60%

Running cost saving vs oil: lower on a standard tariff, higher on a heat pump tariff

26-34 kW

Installed cascade capacity for two twin-fan units at A7/W35

Typical ranges for this class of system, confirmed per project at design stage. Cost comparison based on kerosene at approx. 92p/litre and electricity at 26.11p/kWh, or 13-15p on a heat pump tariff (August 2026).

Where a cascade earns its place: a heat loss above roughly 18 kW, a second heat demand such as a pool or annexe, or a property that cannot afford to be without heat if one appliance faults.

Have a property that asks more than one thing of its heating?

High heat loss, a pool, an annexe or awkward emitters - these are the projects where a standard single-unit specification quietly underperforms. We design cascade and multi-load systems from the heat loss up.

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